1 ELEMENTARY MATHEMATICS

Learning outcomes

By the end of this topic you should be able to:

1.1 Understand the meaning of the **sets of numbers** $\mathbb{N}, \mathbb{Z}, \mathbb{R}, \mathbb{Q}$ . . . 1-1
1.2 Calculate the sum of a sequence of numbers, represented using **sigma notation** . . . 1-3
1.3 **Expand** out brackets and **factorise** quadratics . . . 1-4
1.4 Simplify expressions involving **exponents** . . . 1-6
1.5 Calculate the **logarithm** of a number in a given base . . . 1-7
1.6 **Convert a mixed number** from base b to base 10 . . . 1-8
1.7 Convert an integer from base 10 **to another base** . . . 1-10
1.8 Represent a number in **scientific notation** . . . 1-12
1.9 Solve linear and quadratic **equations** . . . 1-13
1.10 Solve single and double **inequalities** . . . 1-15
1.11 Understand basic definitions relating to **functions** . . . 1-17
1.12 Recognise **graphs** of common functions . . . 1-20
1.13 Recognise and use some **common functions in computing** . . . 1-22

1.1 Understand the meaning of the **sets of numbers** $\mathbb{N}, \mathbb{Z}, \mathbb{R}, \mathbb{Q}$

NATURAL NUMBERS $\mathbb{N}$

The numbers that we use to count, namely $1, 2, 3, \ldots$ are called **natural numbers**. You may have heard such numbers described as 'whole numbers'. The *set*[a] of all natural numbers is denoted by $\mathbb{N}$.

So $\mathbb{N} = \{1, 2, 3, \ldots\}$, where $\ldots$ means the (obvious) pattern continues forever. The numbers $1, 2, 3, \ldots$ are called *members* of the set $\mathbb{N}$.

[a] We will learn more about sets in the next topic.

There is no consensus on whether zero is a natural number or not. In this module, we do not include zero in $\mathbb{N}$. If needed, we will use $\mathbb{N}_0$ to denote the set of natural numbers with zero included.

NOTATION: $\in$, $\notin$

$x \in \mathbb{N}$ means 'x is a member of $\mathbb{N}$'. For example, $4 \in \mathbb{N}$, $0 \in \mathbb{N}_0$.
$x \notin \mathbb{N}$ means 'x is not a member of $\mathbb{N}$'. For example, $-1 \notin \mathbb{N}$, $0 \notin \mathbb{N}$.

INTEGERS $\mathbb{Z}$

The natural numbers along with their negative counterparts and zero are called the **integers**. The set of all integers is denoted by $\mathbb{Z}$.

So $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$. The set of all positive integers is denoted $\mathbb{Z}^+$, and the set of all negative integers is denoted $\mathbb{Z}^-$.

Author: Glenn Hawe © Ulster University 2024

REAL NUMBERS $\mathbb{R}$

Any number which falls anywhere on the number line is a **real number**. The set of all real numbers is denoted by $\mathbb{R}$. The set of all positive real numbers is denoted $\mathbb{R}^+$, and the set of all negative real numbers is denoted $\mathbb{R}^-$.

RATIONALS $\mathbb{Q}$

Real numbers which can be expressed as fractions (integer numerator and integer non-zero denominator) are called the **rational numbers**. The set of all rational numbers is denoted by $\mathbb{Q}$.

So $\mathbb{Q} = \left\{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\right\}$.

Worked Examples

EXAMPLE 1.1

Which of the sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ are the following numbers members of?

1.1.1 7	**Solution** $7 \in \mathbb{Q}, \mathbb{N}, \mathbb{Z}, \mathbb{R}$
1.1.2 -0.1	**Solution** $-0.1 = -\frac{1}{10} \in \mathbb{Q}, \mathbb{R}$. All terminating decimals are rational.
1.1.3 $0.\dot{3} = 0.333\ldots$	**Solution** $0.\dot{3} = \frac{1}{3} \in \mathbb{Q}, \mathbb{R}$. All repeating decimals are rational.

Classroom Exercises

EXERCISE 1.1

Which of the sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ are the following numbers members of?

1.1.1 -7	**Solution** $-7 \in \mathbb{Q}, \mathbb{Z}, \mathbb{R}$
1.1.2 $\sqrt{3}$	**Solution** $\sqrt{3} \in \mathbb{R}$. The square root of a non-square number is irrational.
1.1.3 $0.\dot{6}$	**Solution** $0.\dot{6} = 0.666\ldots = \frac{2}{3} \in \mathbb{Q}, \mathbb{R}$. All repeating decimals are rational.
1.1.4 $0.\dot{9}$	**Solution** $0.\dot{9} = 0.999\ldots = 1 \in \mathbb{Q}, \mathbb{N}, \mathbb{Z}, \mathbb{R}$. Note that $0.\dot{9} = 1$ exactly; see, e.g. Wallis, p.3.

1.2 Calculate the sum of a sequence of numbers, represented using **sigma notation**

SIGMA NOTATION
$\sum_{i=1}^{n} a_i$

Suppose we have a sequence of numbers, $a_1, a_2, a_3, \ldots$.

The 'sigma' notation $\sum_{i=1}^{n} a_i$ or $\sum\limits_{i=1}^{n} a_i$ denotes the following sum:

$$\sum_{i=1}^{n} a_i = a_1 + a_2 + a_3 + \ldots + a_{n-1} + a_n$$

So, for example, $\sum_{i=1}^{3} a_i = a_1 + a_2 + a_3$.

Be careful: The lower limit for the index i may not always be 1.
For example, $\sum_{i=5}^{8} a_i = a_5 + a_6 + a_7 + a_8$.

Worked Examples

EXAMPLE 1.2

Write each of the following expressions as a sum and evaluate it.

1.2.1 $\sum\limits_{i=1}^{5} i$	**Solution** $\sum\limits_{i=1}^{5} i = 1 + 2 + 3 + 4 + 5 = 15$
1.2.2 $\sum\limits_{i=1}^{3} i^2$	**Solution** $\sum\limits_{i=1}^{3} i^2 = 1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$
1.2.3 $\sum\limits_{i=2}^{4} \frac{i}{i+1}$	**Solution** Note that the sum starts at $i = 2$, so $\sum\limits_{i=2}^{4} \frac{i}{i+1} = \frac{2}{3} + \frac{3}{4} + \frac{4}{5} = \frac{133}{60}$

Classroom Exercises

EXERCISE 1.2

Write each of the following expressions as a sum and evaluate it.

1.2.1

$$\sum_{i=1}^{3} i^3$$

Solution

$$\sum_{i=1}^{3} i^3 = \underbrace{1^3}_{i=1} + \underbrace{2^3}_{i=2} + \underbrace{3^3}_{i=3}$$

$$= 1 + 8 + 27$$

$$= 36$$

1.2.2

$$\sum_{i=3}^{6} i(i-2)$$

Solution

$$\sum_{i=3}^{6} i(i-2) = \underbrace{3(3-2)}_{i=3} + \underbrace{4(4-2)}_{i=4} + \underbrace{5(5-2)}_{i=5} + \underbrace{6(6-2)}_{i=6}$$

start at $i=3$

$$= 3(1) + 4(2) + 5(3) + 6(4)$$

$$= 3 + 8 + 15 + 24$$

$$= 50$$

1.2.3

$$\sum_{i=1}^{6} (i^2+1)$$

Solution

$$\sum_{i=1}^{6} (i^2+1) = \underbrace{(1^2+1)}_{i=1} + \underbrace{(2^2+1)}_{i=2} + \underbrace{(3^2+1)}_{i=3}$$

$$+ \underbrace{(4^2+1)}_{i=4} + \underbrace{(5^2+1)}_{i=5} + \underbrace{(6^2+1)}_{i=6}$$

$$= 2 + 5 + 10 + 17 + 26 + 37$$

$$= 97$$

1.3 **Expand** out brackets and **factorise** quadratics

EXPANDING BRACKETS

To multiply one expression by another expression, multiply each term in the first expression by each term in the second expression and sum the resulting terms together. For example:

$$(a+b)(c+d) = ac + ad + bc + bd$$

So, for example consider $(x+3)(x-1)$. Multiplying each term in the first expression $(x+3)$ by each term in the second expression $(x-1)$ gives:

$$(x+3)(x-1) = x^2 - x + 3x - 3 = x^2 + 2x - 3$$

FACTORISING

Factorising is the reverse of expanding brackets: i.e. it is the process of rewriting an expression in terms of the product of its factors.

QUADRATIC EXPRESSION

A **quadratic expression** is an expression containing one or more terms in which the variable's highest power is 2, i.e. it is an expression of the form ax^2+bx+c, where a is the (non-zero) coefficient of x^2, b is the coefficient of $x^1 (= x)$, and c is the coefficient of $x^0 (= 1)$.

FACTORISING $x^2 + bx + c$

To factorise a quadratic expression of the form $x^2 + bx + c$ (note $a = 1$), we must find two integers p, q that have a product of c, and a sum of b. Then $x^2+bx+c = (x+p)(x+q)$ because $(x+p)(x+q) = x^2 + (p+q)x + pq = x^2 + bx + c$.

For example, consider x^2+5x+6. Here $b = 5$ and $c = 6$, so we must find two integers whose product is 6 and whose sum is 5. The only two such integers are 2 and 3, so the factorization is $(x+2)(x+3)$.

Worked Examples

EXAMPLE 1.3

Expand and simplify the following expressions.

1.3.1 $-4(4x-3y)$	**Solution** $-4 \cdot 4x + -4 \cdot -3y = -16x + 12y$
1.3.2 $(x-3)(x-5)$	**Solution** $x \cdot x + x \cdot -5 + -3 \cdot x + -3 \cdot -5 = x^2 - 5x - 3x + 15 = x^2 - 8x + 15$
1.3.3 $(x+y)(x-y)$	**Solution** $x \cdot x + x \cdot -y + y \cdot x + y \cdot -y = x^2 - xy + xy - y^2 = x^2 - y^2$

EXAMPLE 1.4

Factorise the following expressions.

1.4.1 $x^2 + 6x + 9$	**Solution** Two integers whose product is 9 and whose sum is 6 are 3 and 3. So $x^2 + 6x + 9 = (x + 3)(x + 3)$.
1.4.2 $x^2 + 3x - 4$	**Solution** Two integers whose product is -4 and whose sum is 3 are 4 and -1. So $x^2 + 3x - 4 = (x + 4)(x - 1)$.
1.4.3 $x^2 - 7x + 12$	**Solution** Two integers whose product is 12 and whose sum is -7 are -3 and -4. So $x^2 - 7x + 12 = (x - 3)(x - 4)$.

1.4 Simplify expressions involving **exponents**

BASE AND EXPONENT

If n is a natural number, then the expression a^n means $a \times a \times a \times \ldots \times a$ where there are n occurrences of a in the product.
The number a is referred to as the **base** and the number n is referred to as the **exponent** (sometimes also called the *index* or *power*).

For example, $2^3 = 2 \times 2 \times 2$ and $3^5 = 3 \times 3 \times 3 \times 3 \times 3$

Negative exponents are defined as follows: $a^{-n} = \dfrac{1}{a^n}$

$a^0 = 1$ for all $a \neq 0$.

$a^m \times a^n = a^{m+n}$

When expressions with the same base are multiplied, the indices are added: $a^m \times a^n = a^{m+n}$

$\dfrac{a^m}{a^n} = a^{m-n}$

When expressions with the same base are divided, the indices are subtracted: $\dfrac{a^m}{a^n} = a^{m-n}$

$(a^m)^n = a^{mn}$

$(a^m)^n = a^{mn}$

It follows from (1) above that $a^{\frac{1}{2}} a^{\frac{1}{2}} = a^1 = a$, so $a^{\frac{1}{2}}$ is the square root of a, i.e. $a^{\frac{1}{2}} = \sqrt{a}$.

In general, $a^{\frac{1}{n}} = \sqrt[n]{a}$.

Worked Examples

EXAMPLE 1.5

Simplify each of the following, writing your answer using positive exponents only:

1.5.1 $3^4 \times 3^5$	**Solution** $3^4 \times 3^5 = 3^{(4+5)} = 3^9$
1.5.2 $\frac{x^6}{x^2}$	**Solution** $\frac{x^6}{x^2} = x^{6-2} = x^4$
1.5.3 $(y^5)^3$	**Solution** $(y^5)^3 = y^{5\times 3} = y^{15}$
1.5.4 $3^3 \times 4^5$	**Solution** No rule is applicable: cannot be simplified.

Classroom Exercises

EXERCISE 1.3

Simplify each of the following, writing your answer using positive exponents only:

1.3.1 $5y^2z^{-3}$	**Solution** $5y^2z^{-3} = \frac{5y^2}{z^3}$
1.3.2 $(2xy)^{-2}$	**Solution** $(2xy)^{-2} = \frac{1}{(2xy)^2} = \frac{1}{2^2x^2y^2} = \frac{1}{4x^2y^2}$
1.3.3 t^3t^{-3}	**Solution** $t^3t^{-3} = t^{3-3} = t^0 = 1$ (provided $t \neq 0$)
1.3.4 $\frac{(xy)^3}{xy^2}$	**Solution** $\frac{(xy)^3}{xy^2} = \frac{x^3y^3}{xy^2} = x^{3-1}y^{3-2} = x^2y$

1.5 Calculate the **logarithm** of a number in a given base

LOGARITHM

The process of taking powers can be inverted using logarithms.

The **logarithm** of x with respect to base b, written as $\log_b x$, is defined to be the number y such that $b^y = x$.

Logarithms are not defined for base 1, as $1^y = 1$ for any value of y.

Worked Examples

EXAMPLE 1.6

What are the values of the following?

1.6.1 $\log_2 8$	**Solution** $2^3 = 8$, so $\log_2 8 = 3$
1.6.2 $\log_{16} 4$	**Solution** $16^{0.5} = 4$, so $\log_{16} 4 = 0.5$
1.6.3 $\log_3 9$	**Solution** $3^2 = 9$, so $\log_3 9 = 2$
1.6.4 $\log_4 \frac{1}{2}$	**Solution** $4^{-0.5} = \frac{1}{4^{0.5}} = \frac{1}{2}$, so $\log_4 \frac{1}{2} = -0.5$

Classroom Exercises

EXERCISE 1.4

What are the values of the following?

1.4.1 $\log_4 4$	**Solution** $4^1 = 4$ so $\log_4 4 = 1$
1.4.2 $\log_{27} 3$	**Solution** $27^{\frac{1}{3}} = \sqrt[3]{27} = 3$ so $\log_{27} 3 = \frac{1}{3}$
1.4.3 $\log_{10} 0.1$	**Solution** $10^{-1} = \frac{1}{10} = 0.1$ so $\log_{10} 0.1 = -1$
1.4.4 $\log_2 \frac{1}{4}$	**Solution** $2^{-1} = \frac{1}{2^2} = \frac{1}{4}$ so $\log_2 \frac{1}{4} = -2$

1.6 **Convert a mixed number** from base *b* to base 10

We are used to writing numbers in base 10 and interpreting a number as a sum of successive powers of ten. For example: 385 means 3 times 10^2, plus 8 times 10^1, plus 5 (times $10^0 = 1$)

When we write a number $a_n \dots a_2 a_1 a_0 . c_1 c_2 \dots c_m$ (in base 10), it means:

$$\sum_{i=0}^{n} a_i . 10^i + \sum_{j=1}^{m} c_j . 10^{-j}$$

There is no special reason to use base 10: we can use any base we want. In computer science, base 2 is commonly used. Numbers written in base 2 are called **binary** numbers.

When writing numbers in a base other than 10, it is common to explicitly write down the base as a suffix, e.g. $(132)_4$ is a number in base 4.

Note: the number of the base is written in base 10. If the number of the base was written in the actual base, then it would always be 10 (one - zero). e.g. if we were to write 4 in base 4, then it is written as 10.

In base 10 we need ten digits to represent numbers. Generally, in base b, we need b digits. For example, in base 2, we only need two digits: 0 and 1.

If the base b is greater than 10, then we need new symbols to represent digits. For example, in base 16 (called hexadecimal), we use A-F to represent numbers 10-15.

Converting a number from base b to base 10

If a number $a_n \dots a_2 a_1 a_0 . c_1 c_2 \dots c_m$ is in base b, then we may convert it to base 10 as follows:

$$\sum_{i=0}^{n} a_i . b^i + \sum_{j=1}^{m} c_j . b^{-j}$$

Worked Examples

EXAMPLE 1.7

Convert the following to base 10:

1.7.1 $(104)_5$	**Solution** $(104)_5 = 4(5^0) + 0(5^1) + 1(5^2) = 4(1) + 0(5) + 1(25) = 29$
1.7.2 $(253)_7$	**Solution** $(253)_7 = 3(7^0) + 5(7^1) + 2(7^2) = 3(1) + 5(7) + 2(49) = 136$
1.7.3 $(0.01)_2$	**Solution** $(0.01)_2 = 0(2^{-1}) + 1(2^{-2}) = 0(0.5) + 1(0.25) = 0.25$
1.7.4 $(101.31)_6$	**Solution** $(101.31)_6 = 1(6^0)+0(6^1)+1(6^2)+3(6^{-1})+1(6^{-2})$ $= 1(1) + 0(6) + 1(36) + 3(\frac{1}{6}) + 1(\frac{1}{6^2})$ $= 37\frac{19}{36} = 37.52\dot{7}$

Classroom Exercises

EXERCISE 1.5

Calculate the value (in base 10) of each of the following:

1.5.1 $(112)_3$	**Solution** $(112)_3 = 2(3^0) + 1(3^1) + 1(3^2) = 2(1) + 1(3) + 1(9) = 14$
1.5.2 $(1B)_{16}$	**Solution** $(1B)_16 = 11(16^0) + 1(16^1) = 11 + 16 = 27$
1.5.3 $(210.04)_8$	**Solution** $(210)_8 = 0(8^0)+1(8^1)+2(8^2) = 0+8+128 = 136$ $(0.04)_8 = 0(8^{-1}) + 4(8^{-2}) = \frac{4}{64} = \frac{1}{16}$. So overall $(210.04)_8 = 136\frac{1}{16}$
1.5.4 $(22.6)_7$	**Solution** $(22)_7 = 2(7^0) + 2(7^1) = 2 + 14 = 16$. $(0.6)_7 = 6(7^{-1}) = \frac{6}{7}$. So overall $(22.6)_7 = 16\frac{6}{7}$

1.7 Convert an integer from base 10 to another base

To convert an integer from base 10 to another base, we use the following method of division.

Example: Convert 108 to base 7

Repeatedly divide 7 (the target base) into 108 (the integer we are converting):

7	108	
7	15	+3
7	2	+ 1
	0	+ 2

Reading the remainders from bottom to top gives us 108 in base 7. i.e. $108 = (213)_7$

Classroom Exercises

EXERCISE 1.6

Convert the following numbers from base 10 to the given base:

1.6.1

126 to base 2

Solution

$$\begin{array}{r|l} 2 & 126 \\ 2 & 63 + 0 \\ 2 & 31 + 1 \\ 2 & 15 + 1 \\ 2 & 7 + 1 \\ 2 & 3 + 1 \\ 2 & 1 + 1 \\ & 0 + 1 \end{array}$$

representation in base 2 is

$$\begin{array}{ccccccc} 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 64 & 32 & 16 & 8 & 4 & 2 & 1 \end{array}$$

$+$

$= 126$

1.6.2

54 to base 5

Solution

$$\begin{array}{r|l} 5 & 54 \\ 5 & 10 + 4 \\ 5 & 2 + 0 \\ & 0 + 2 \end{array}$$

representation in base 5 is

$$\begin{array}{ccc} 5^2 & 5^1 & 5^0 \\ 2 & 0 & 4 \end{array}$$

$4(5^0) + 0(5^1) + 2(5^2) = 54$

1.6.3

103 to base 6

Solution

$$\begin{array}{r|l} 6 & 103 \\ 6 & 17 + 1 \\ 6 & 2 + 5 \\ & 0 + 2 \end{array}$$

representation in base 6 is

$$\begin{array}{ccc} 6^2 & 6^1 & 6^0 \\ 2 & 5 & 1 \end{array}$$

$1(6^0) + 5(6^1) + 2(6^2)$

$= 1 + 30 + 72 = 103$

1.8 Represent a number in **scientific notation**

SCIENTIFIC NOTATION

Very large numbers and very small numbers are cumbersome to write out fully in the way we would normally write numbers. For example, the mass of the Earth is approximately 5972200000000000000000000 kg. It is more convenient to write large numbers like this in **scientific notation**.

MANTISSA

In scientific notation, a number is written in the form $m \times 10^n$, where the real number m is called the **mantissa** and the integer n is called the exponent. In normalized notation, the absolute value of the mantissa satisfies $1 \leq |m| < 10$.

To write a number in scientific notation, move the decimal point so that there is one digit (which cannot be zero) before the decimal point. This gives you the mantissa m. One you have the mantissa, the value of the exponent is the number of places you need to move the decimal point to get the original number.

For example, the weight of the Earth given above, written in scientific notation, is 5.9722×10^{24} kg. The mantissa is 5.9722, and we need to move the decimal point 24 places to the right to get the original number, so the exponent is 24. (If the decimal point in the mantissa needs to move to the left to get the original number, then the exponent is negative).

Worked Examples

EXAMPLE 1.8

Write the following numbers in scientific notation:

1.8.1 6200	**Solution** $6200 = 6.2 \times 10^3$
1.8.2 −1450000	**Solution** $-1450000 = -1.45 \times 10^6$
1.8.3 0.00097	**Solution** $0.00097 = 9.7 \times 10^{-4}$
1.8.4 −0.0000013	**Solution** $-0.0000013 = -1.3 \times 10^{-6}$

EXAMPLE 1.9

Write the following numbers out fully:

1.9.1 3.3×10^{-4}	**Solution** The exponent is -4, so move the decimal point 4 places left: 0.00033
1.9.2 5.6278×10^{2}	**Solution** The exponent is 2, so move the decimal point 2 places right: 562.78
1.9.3 -9.5×10^{6}	**Solution** The exponent is 6, so move the decimal point 6 places right: −9500000

1.9 Solve linear and quadratic **equations**

A linear equation involving one variable x is of the form $ax + b = c$, and may be solved by rearranging the equation so that it is in the form $x = \ldots$

For example, to solve $3x + 4 = 10$:
subtract 4 from both sides: $3x = 6$ divide through both sides by 3: $x = 2$

Some linear equations may not have a solution, e.g. $x = x + 1$.
Some linear equations may have infinitely many solutions, e.g. $2x + 3 = 2x + 3$

A quadratic equation involving one variable x is usually written in the form $ax^2 + bx + c = 0$. One way to solve this is using the formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

DISCRIMINANT

In the formula above, the term $b^2 - 4ac$ is called the **discriminant**. The formula involves taking the square root of this number, so the sign of the discriminat determines how many real valued solutions exist to the quadratic equation.
If $b^2 - 4ac < 0$: the equation has no real-valued solutions
If $b^2 - 4ac = 0$: the equation has one real-valued solution
If $b^2 - 4ac > 0$: the equation has two distinct real-valued solutions

Worked Examples

EXAMPLE 1.10

Solve the following quadratic equations:

1.10.1

$2x^2 + 7x - 4 = 0$

Solution

Using $a = 2, b = 7, c = -4$ in the formula:

$$\begin{aligned} x &= \frac{-7 \pm \sqrt{7^2 - 4(2)(-4)}}{2(2)} \\ &= \frac{-7 \pm \sqrt{49 - (-32)}}{4} \\ &= \frac{-7 \pm \sqrt{81}}{4} \\ &= \frac{-7 \pm 9}{4} \end{aligned}$$

So the solutions are:
$x = \frac{-16}{4} = -4$ and $x = \frac{2}{4} = 0.5$

1.10.2

$x^2 - 8x + 16 = 0$

Solution

Using $a = 1, b = -8, c = 16$ in the formula:

$$\begin{aligned} x &= \frac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(16)}}{2(1)} \\ &= \frac{8 \pm \sqrt{64 - 64}}{2} \\ &= \frac{8 \pm \sqrt{0}}{4} \\ &= 2 \end{aligned}$$

So the single solution is $x = 2$

Classroom Exercises

EXERCISE 1.7

Solve the following quadratic equations:

1.7.1

$x^2 - 4x - 21 = 0$

Solution

Using $a = 1, b = -4, c = -21$ in the formula:

$$\begin{aligned} x &= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(-21)}}{2(1)} \\ &= \frac{4 \pm \sqrt{16 - (-84)}}{2} \\ &= \frac{4 \pm \sqrt{100}}{2} \\ &= \frac{4 \pm 10}{2} \end{aligned}$$

So the solutions are:
$x = \frac{-6}{2} = -3$ and $x = \frac{14}{2} = 7$

1.7.2

$x^2 - 6x + 9 = 0$

Solution

Using $a = 1, b = -6, c = 9$ in the formula:

$$\begin{aligned} x &= \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(9)}}{2(1)} \\ &= \frac{6 \pm \sqrt{36 - 36}}{2} \\ &= \frac{6 \pm \sqrt{0}}{2} \\ &= \frac{6}{2} = 3 \end{aligned}$$

So the single solution is $x = 3$.

1.10 Solve single and double **inequalities**

RULES TO MANIPULATE INEQUALITIIES

Suppose we have the inequality $a < b$. Then we can manipulate it as follows:

$a + c < b + c$ for any c

$ca < cb$ if c is positive

$ca > cb$ if c is negative

SINGLE INEQUALITY

To solve a single inequality, manipulate the inequality so that you have $x < \ldots$ (or $x > \ldots$, or similar)

For example, to solve $2x + 16 < 4x + 28$:
Subtract 2x from both sides: $16 < 2x + 28$
Subtract 28 from both sides: $-12 < 2x$
Divide both sides by 2: $-6 < x$.
Therefore the solution to $2x + 16 < 4x + 28$ is $x > -6$ (i.e. all values of x greater than -6).

DOUBLE INEQUALITY

An example of a **double inequality** is $2x - 5 < 5x - 2 < 3x + 4$
To solve a double inequality, you must rewrite it as two single inequalities, solve both of those and then combine the answers.

For example, to solve $2x - 5 < 5x - 2 < 3x + 4$
Considering first $2x - 5 < 5x - 2$, solving we get $x > -1$
Considering next $5x - 2 < 3x + 4$, solving we get $x < 3$
Combining both answers, we get: $-1 < x < 3$.

Worked Examples

EXAMPLE 1.11

Solve the following inequalities:

1.11.1 $5x - 3 > x + 5$	**Solution** Subtract x from each side: $4x - 3 > 5$ Add 3 to each side: $4x > 8$ Divide both sides by 4: $x > 2$
1.11.2 $-8x < 32$	**Solution** Divide each side by -8: $x > -4$ (note change of inequality sign)
1.11.3 $2x < 7x + 15$	**Solution** Subtract $2x$ from each side: $0 < 5x + 15$ Subtract 15 from each side: $-15 < 5x$ Divide each side by 5: $-3 < x$
1.11.4 $2x + 12 < 7x - 3 < 5x + 21$	**Solution** Consider $2x + 12 < 7x - 3$ first. Subtract $2x$ from each side: $12 < 5x - 3$ Add 3 to each side: $15 < 5x$ Divide each side by 5: $3 < x$ Now consider $7x - 3 < 5x + 21$ Subtract $5x$ from each side: $2x - 3 < 21$ Add 3 to each side: $2x < 24$, so $x < 12$ So overall, $3 < x < 12$

1.11 Understand basic definitions relating to **functions**

MAPPING

A **mapping** is a process that takes an input value x and gives an output value $f(x)$.

IMAGE

f(x) is called the **image** of x.

If each input value comes from a set A, and each output value belongs to a set B, then we represent the mapping f as $f: A \to B$.

DOMAIN, CODOMAIN

The set A of input values is called the **domain**.

The set B is called the **codomain**. Every image value $f(x)$ belongs to B, although not every value in B may be the image of some input.

RANGE

The set of all image values alone is called the **range**.

FUNCTION

A **function** is a mapping in which each input value from the domain has one and only one image in the codomain.

If there is an input value in the domain that has two distinct images, then the mapping is **not** a function.

There are four basic types of mapping, explained below.

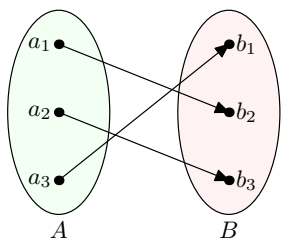
One-to-one and many-to-one mappings are functions.

Many-to-one and many-to-many mappings are not functions.

ONE-TO-ONE MAPPING

A **one-to-one** mapping maps each input x in the domain to a distinct output $f(x)$ in the codomain. No member of the codomain is the image of more than one input.

Example of discrete one-to-one mapping

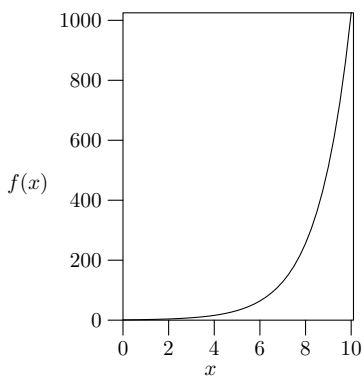


The domain is the set $A = \{a_1, a_2, a_3\}$.

The codomain is the set $B = \{b_1, b_2, b_3\}$.

The range is the set $B = \{b_1, b_2, b_3\}$.

Example of continuous one-to-one mapping



$f: \mathbb{R}_0^+ \to \mathbb{R}$ where $f(x) = 2^x$

The domain is $\mathbb{R}_0^+$ (the set of positive real numbers including zero).

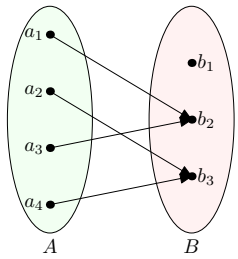
The codomain is $\mathbb{R}$.

The range is the set of real numbers greater than or equal to 1 (as $2^0 = 1$ is the smallest output value).

MANY-TO-ONE MAPPING

In a **many-to-one** mapping, each input x from the domain is mapped to one single image $f(x)$ in the codomain. Different inputs may have the same image, i.e. it is possible for $x_1 \neq x_2$ and $f(x_1) = f(x_2)$.

Example of discrete many-to-one mapping

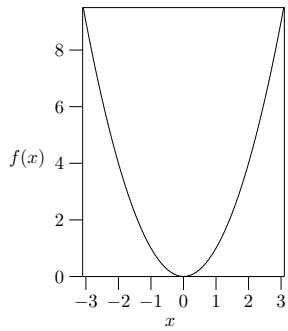


The domain is the set $A = \{a_1, a_2, a_3, a_4\}$.

The codomain is the set $B = \{b_1, b_2, b_3\}$.

The range is the set $\{b_2, b_3\}$.

Example of continuous many-to-one mapping



$f\colon \mathbb{R} \to \mathbb{R}$ where $f(x) = x^2$.

The domain is $\mathbb{R}$.

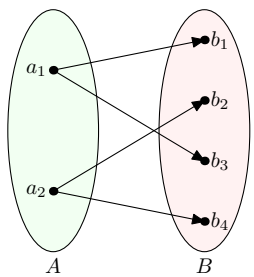
The codomain is $\mathbb{R}$.

The range is $\mathbb{R}_0^+$ ($0^2 = 0$ is the smallest output value).

ONE-TO-MANY MAPPING

In a **one-to-many** mapping, an input x from the domain may be mapped to multiple distinct images in the codomain. Each value in the codomain is the image of at most one point in the domain.

Example of discrete one-to-many mapping

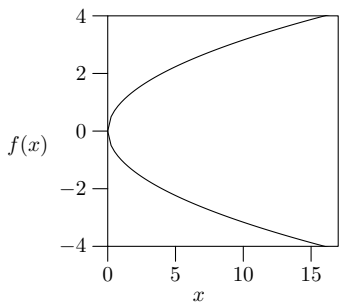


The domain is the set $A = \{a_1, a_2\}$.

The codomain is the set $B = \{b_1, b_2, b_3, b_4\}$.

The range is the set $\{b_1, b_2, b_3, b_4\}$.

Example of continuous one-to-many mapping



$f\colon \mathbb{R}_0^+ \to \mathbb{R}$ where $f(x) = \pm\sqrt{x}$.

The domain is $\mathbb{R}_0^+$.

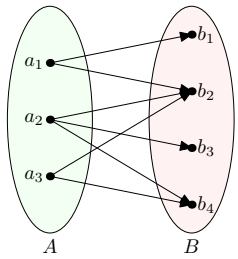
The codomain is $\mathbb{R}$.

The range is $\mathbb{R}$.

MANY-TO-MANY MAPPING

In a **many-to-many** mapping, an input x from the domain may be mapped to multiple distinct images in the codomain. Each value in the codomain may be the image of multiple distinct points in the domain.

Example of discrete many-to-many mapping

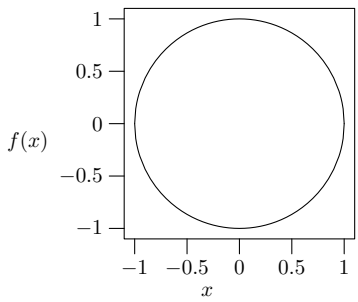


The domain is the set $A = \{a_1, a_2, a_3\}$.

The codomain is the set $B = \{b_1, b_2, b_3, b_4\}$.

The range is the set $\{b_1, b_2, b_3, b_4\}$.

Example of continuous many-to-many mapping



$f\colon [-1, 1] \to \mathbb{R}$

where $f(x) = \pm\sqrt{1 - x^2}$.

The domain is the set of real numbers x where $-1 \le x \le 1$.

The codomain is $\mathbb{R}$.

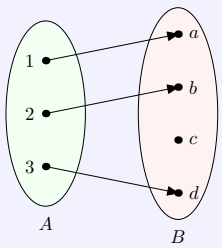
The range is the set of real numbers y where $-1 \le y \le 1$.

Worked Examples

EXAMPLE 1.12

For each of the following mappings, determine what the domain, codomain and range is. State what type of mapping it represents, and whether it is a function or not.

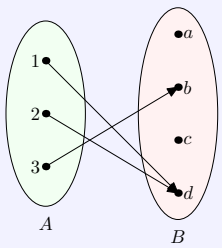
1.12.1



Solution

The domain is $\{1, 2, 3\} = A$.
The codomain is $B = \{a, b, c, d\}$.
The range is $\{a, b, d\}$.
This is a one-to-one mapping.
It is a function.

1.12.2



Solution

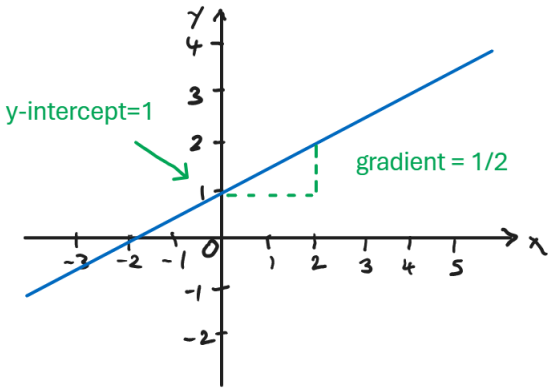
The domain is $\{1, 2, 3\} = A$.
The codomain is $B = \{a, b, c, d\}$.
The range is $\{b, d\}$.
This is a many-to-one mapping.
It is a function.

1.12 Recognise **graphs** of common functions

GRAPH OF STRAIGHT-LINE

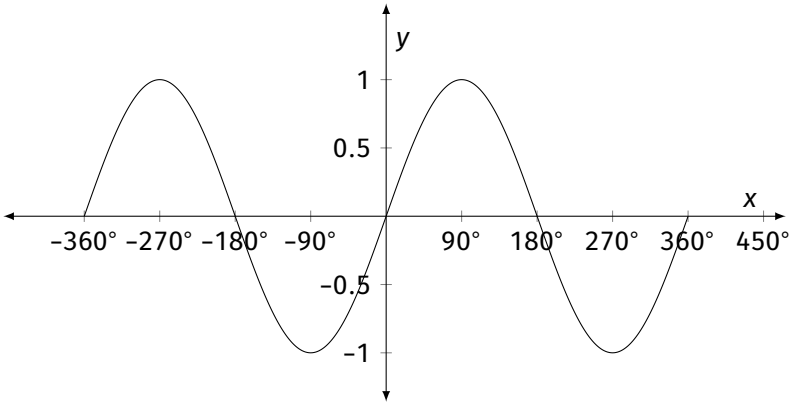
A straight line has the equation $y = mx + c$, where m is the gradient of the line (how steep the slope is), and c is the value of where the line intercepts the y-axis (i.e. where $x = 0$).

Below is the graph of the straight line $y = \frac{1}{2}x + 1$.

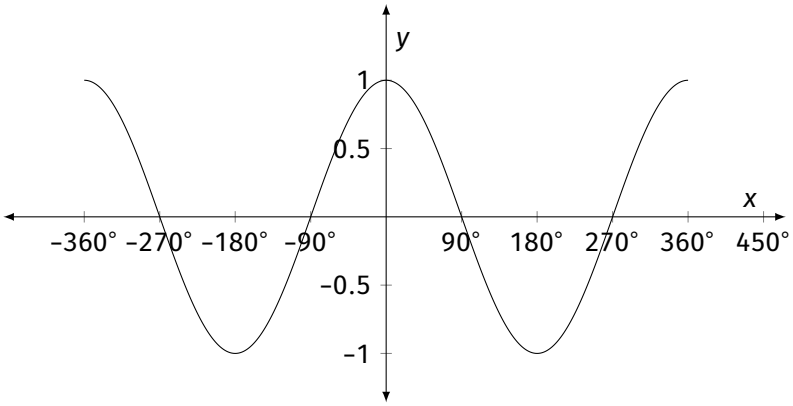


Graphs of trignometric functions

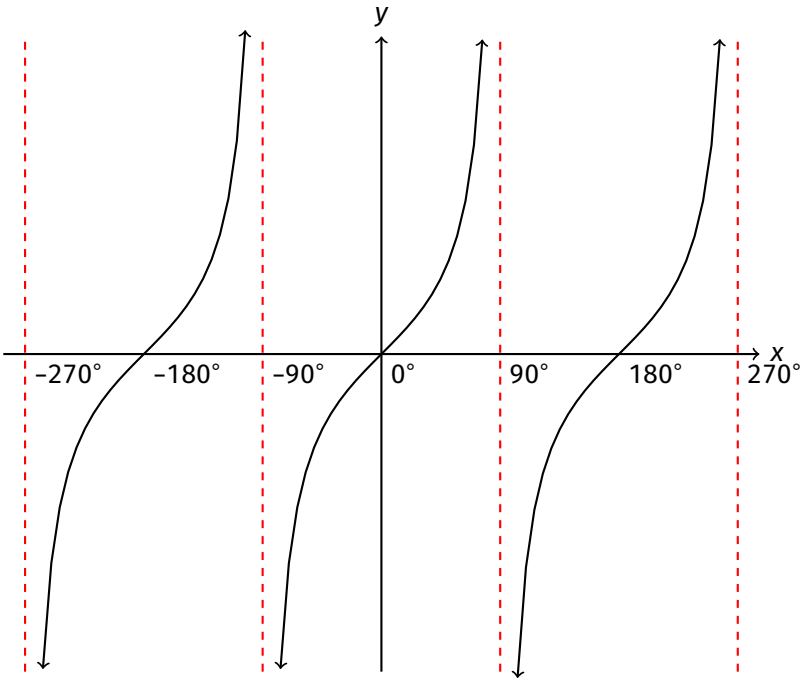
The three basic trignometric functions are $\sin(x), \cos(x)$ and $\tan(x)$.
The graph of $\sin(x)$ is as follows:



The graph of $\cos(x)$ is as follows:



The graph of $\tan(x)$ is as follows:



1.13 Recognise and use some **common functions in computing**

FLOOR

The **floor** function $\texttt{floor}\colon \mathbb{R} \to \mathbb{Z}$ returns the greatest integer that does not exceed $x \in \mathbb{R}$:

$$\texttt{floor}(x) = \lfloor x \rfloor = \max\{z \in \mathbb{Z} : z \leq x\}$$

CEILING

The **ceiling** function $\texttt{ceil}\colon \mathbb{R} \to \mathbb{Z}$ returns the least integer that is not less than $x \in \mathbb{R}$:

$$\texttt{ceil}(x) = \lceil x \rceil = \min\{z \in \mathbb{Z} : z \geq x\}$$

If x is itself an integer, then $x = \lfloor x \rfloor = \lceil x \rceil$; otherwise $\lceil x \rceil = \lfloor x \rfloor + 1$

INTEGER FUNCTION

The **integer** function $\texttt{int}\colon \mathbb{R} \to \mathbb{Z}$ returns the integer part of $x \in \mathbb{R}$. It is related to the floor and ceiling functions as follows:

$$\texttt{int}(x) = \begin{cases} \lfloor x \rfloor & \text{if } x \geq 0, \\ \lceil x \rceil & \text{if } x < 0. \end{cases}$$

MODULUS FUNCTION

The **modulus** function returns the non-negative value of $x \in \mathbb{R}$ (i.e. the value of x ignoring its sign), aka the **absolute value** of x:

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

SIGN FUNCTION

The **sign** function $\texttt{sgn}\colon \mathbb{R} \to \{-1, 0, 1\}$ extracts the sign of $x \in \mathbb{R}$:

$$\texttt{sgn}(x) = \begin{cases} +1 & \text{if } x > 0, \\ 0 & \text{if } x = 0, \\ -1 & \text{if } x < 0 \end{cases}$$

FACTORIAL

The **factorial** function is defined on $\mathbb{N}_0 = \mathbb{N} \cup \{0\} = \{0, 1, 2, 3, \ldots\}$. '$n$ factorial' is denoted by $n!$ and is defined recursively as follows:

$$n! = \begin{cases} 1 & \text{if } n = 0, \\ n \times (n-1)! & \text{if } n > 1. \end{cases}$$

So, for example:

$$\begin{aligned} 1! &= 1 \times 0! = 1 \times 1 = 1 \\ 2! &= 2 \times 1! = 2 \times 1 = 2 \\ 3! &= 3 \times 2! = 3 \times 2 \times 1 = 6 \\ 4! &= 4 \times 3! = 4 \times 3 \times 2 \times 1 = 24 \\ 5! &= 5 \times 4! = 5 \times 4 \times 3 \times 2 \times 1 = 120 \end{aligned}$$

As we can see $n! = n \times (n-1) \times (n-2) \times \ldots 3 \times 2 \times 1$

Worked Examples

EXAMPLE 1.13

Calculate the floor and ceiling of the following numbers:

1.13.1 8	**Solution** $\lfloor 8 \rfloor = 8$ and $\lceil 8 \rceil = 8$
1.13.2 -3.5	**Solution** $\lfloor -3.5 \rfloor = -4$ and $\lceil -3.5 \rceil = -3$

EXAMPLE 1.14

Calculate `int`(x) for the following values of x:

1.14.1 -3.9	**Solution** $\texttt{int}(-3.9) = -3$
1.14.2 -3.1	**Solution** $\texttt{int}(-3.1) = -3$
1.14.3 4.9	**Solution** $\texttt{int}(4.9) = 4$

EXAMPLE 1.15

Calculate the values of:

1.15.1 $\lvert -3.5 \rvert$	**Solution** 3.5
1.15.2 $-\lvert 3.5 \rvert$	**Solution** -3.5
1.15.3 $-\lvert -3.5 \rvert$	**Solution** -3.5
1.15.4 $\lvert -\lvert -3.5 \rvert \rvert$	**Solution** 3.5

EXAMPLE 1.16

Rewrite the following products in terms of factorials:

1.16.1 $6 \times 5 \times 4 \times 3 \times 2 \times 1$	**Solution** $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 6!$
1.16.2 $15 \times 14 \times 13$	**Solution** $15 \times 14 \times 13 = \frac{15!}{12!}$

Classroom Exercises

EXERCISE 1.8

Calculate the floor and ceiling of the following numbers:

1.8.1

4.1

Solution

$\lfloor 4.1 \rfloor = 4$ and $\lceil 4.1 \rceil = 5$

1.8.2

-6

Solution

$\lfloor -6 \rfloor = -6$ and $\lceil -6 \rceil = -6$

EXERCISE 1.9

Calculate `int`(x) for the following values of x:

1.9.1

-0.6

Solution

$\texttt{int}(-0.6) = 0$

1.9.2

0.2

Solution

$\texttt{int}(0.2) = 0$

1.9.3

7

Solution

$\texttt{int}(7) = 7$

EXERCISE 1.10

Calculate the values of:

1.10.1

$|4.3| + |-0.8|$

Solution

$|4.3| + |-0.8| = 4.3 + 0.8 = 5.1$

1.10.2

$-|-2.2|$

Solution

-2.2

1.10.3

$-|-|-0.5||$

Solution

-0.5

1.10.4

$|-1| - |1|$

Solution

$|-1| - |1| = 1 - 1 = 0$

EXERCISE 1.11

Calculate the value of the following:

1.11.1

$\frac{20!}{19!}$

Solution

20

1.11.2

$3!!$

Solution

$3!! = (3 \times 2 \times 1)! = 6! = 720$

Further resources

Text books

- Chapter 1 of W. D. Wallis (2012), *A Beginner's Guide to Discrete Mathematics*, 2nd ed, Birkhäuser.

- Chapter 2 of J. P. Fortney (2020), *Discrete Mathematics for Computer Science*, CRC Press.

Youtube videos

Link	Learning outcome												
	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10	1.11	1.12	1.13
rootmath	x												
Khan Academy		x											
Maths Kitchen			x										
PatrickJMT				x									
Khan Academy					x								
HCCMathHelp						x							
HCCMathHelp							x						
Tyler DeWitt								x					
The Organic Chemistry Tutor									x				
Brian McLogan										x			
Infinity Learn NEET											x		
Maths Genie												x	
Dragonfly Statistics													x